Student academic dropout risk early warning system

Team number 69

Members

1) **Rudransh Sharma** – Documentation lead
2) **Gandrajupalli Jathin Bharadwaj** – Execution lead
3) **Purvi Jain** – PPT lead
4) **Divyansh Kumar** – Video lead

Repo link – https://github.com/Group-69-ai-proj/AI-model-

What and why?

01

Problem statement (what)

Predicting whether an enrolled higher-education student will Graduate, Drop out or stay Enrolled, from demographic data, tuition fee status, scholarship holding and curricular grades. A predicted Dropout marks a student at high risk.

02

Why

If we can predict which students are at higher risk of dropping out, we can take measures to help them finish their education. This will help reduce dropout rates.

03

Dataset source: Kaggle

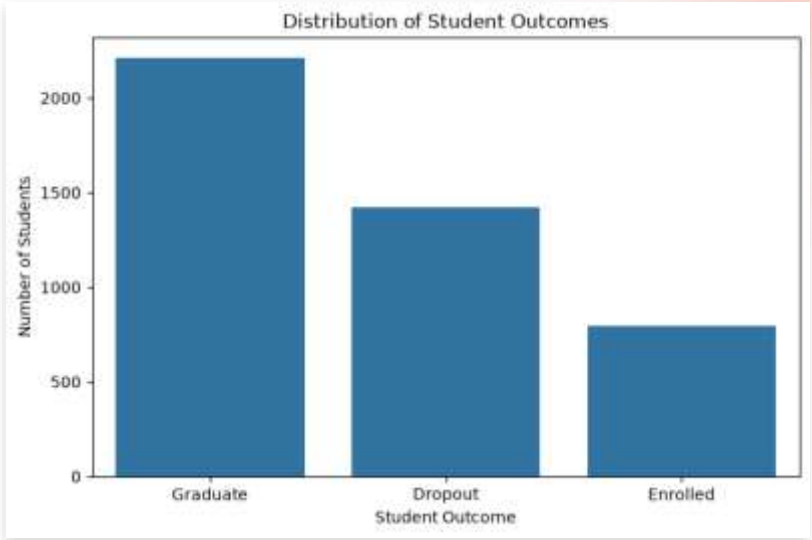
kaggle.com/datasets/thedevastator/higher-education-predictors-of-student-retention

Same data as the UCI repository (Realinho et al., 2022)

04

Rows: 4,424

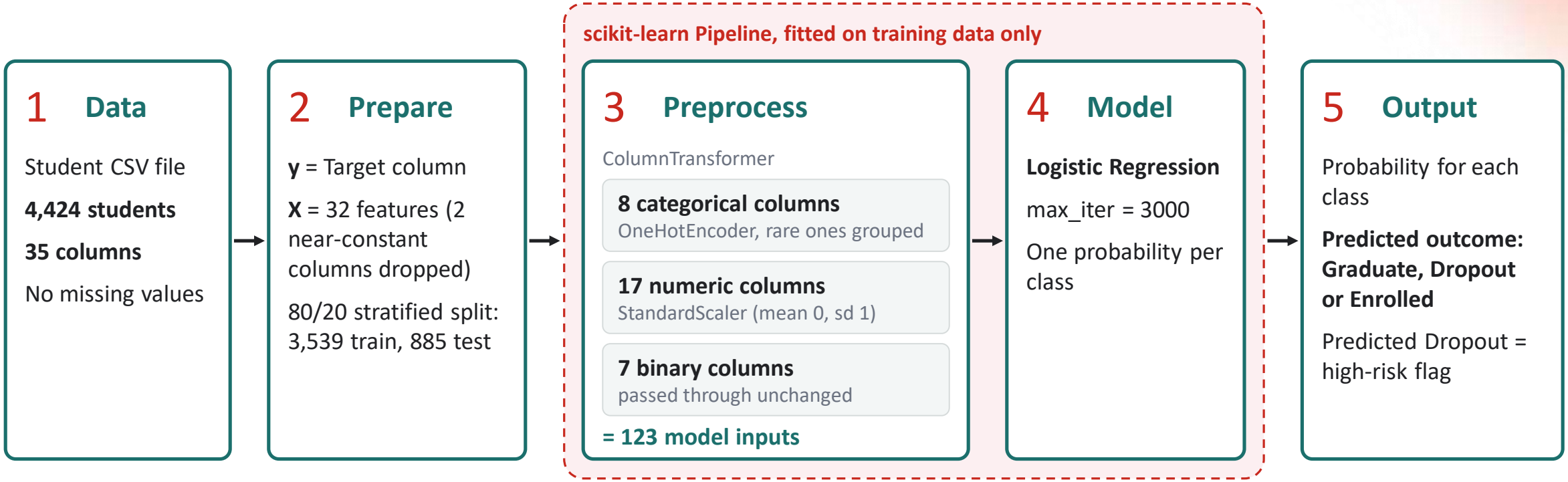
Columns: 35 (no missing values)



From our notebook: 2,209 Graduate, 1,421 Dropout, 794 Enrolled

Fun fact: 87% of students with unpaid tuition fees dropped out, against 25% of those with fees up to date.

Approach: data → model → output



Libraries used

1) NumPy
arrays and maths

2) Pandas
load and summarise data

3) Matplotlib
plots

4) Seaborn
plots by outcome

5) scikit-learn
pipeline, model, metrics

77% right, but “Enrolled” students are hard to spot

Key numbers

Metric	Value
Test set	885 students (20%)
Accuracy	**77.2%**
Macro F1	0.70
Weighted F1	0.76
Baseline: guess Graduate	about 50%

Class	Precision	Recall	F1	Students
Dropout	0.79	0.76	0.77	284
Enrolled	0.56	**0.40**	0.46	159
Graduate	0.81	0.91	0.86	442

Dropout is found well. Enrolled sits between Dropout and Graduate, so only 4 in 10 are caught.

Confusion matrix

actual ↓ predicted →	Dropout	Enrolled	Graduate
Dropout	216	30	38
Enrolled	40	63	56
Graduate	18	20	404

Demo: output of our notebook

```
Accuracy 0.7717514124293785
Macro F1: 0.6998011852995
Weighted F1: 0.7695680942375802

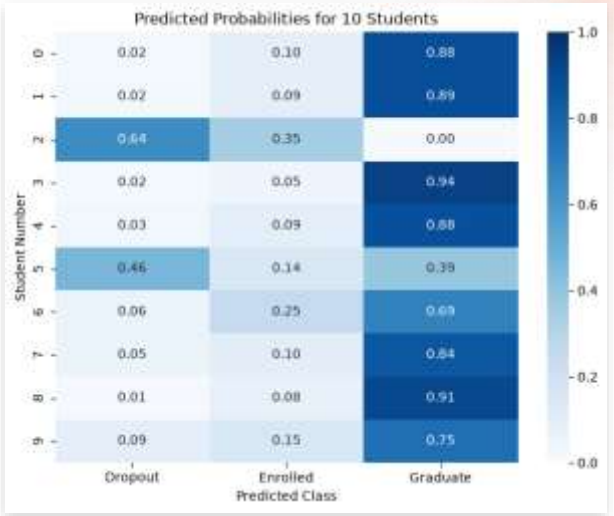
Confusion matrix of a 3 class system
[[216  30  38]
 [ 40  63  56]
 [ 18  20 404]]

Final classification report including precision,recall and f1-score for each class
              precision    recall  f1-score   support

     Dropout       0.79      0.76      0.77       284
    Enrolled       0.56      0.40      0.46       159
    Graduate       0.81      0.91      0.86       442

    accuracy                           0.77       885
   macro avg       0.72      0.69      0.70       885
weighted avg       0.76      0.77      0.76       885
```

Predicted probabilities



Student 5 is a close call: 46% Dropout vs 39% Graduate. Exactly who a university should check.

What we learned

What worked

- A single Pipeline (preprocessing + model), so scaling and encoding are learned only from training data
- Three feature types handled differently: one-hot for codes, scaling for numbers, passthrough for 0/1 columns
- min_frequency=10 groups very rare categories (some columns have up to 46)
- Dropped two almost-constant columns (Nacionality, Educational special needs)
- Stratified split, and macro F1 with per-class scores, not just accuracy

What didn’t work / limitations

- Enrolled is hard to identify (recall 0.40): it is the smallest class and sits between Dropout and Graduate
- Tried class_weight='balanced': Enrolled recall rose from 0.40 to 0.64 (F1 0.46 → 0.50), but accuracy fell from 77% to 72%, so the trade-off wasn’t worth it
- Only one model and one train/test split were tried, so results may vary a bit with another split
- If we had more time: compare a more flexible model, and use cross-validation

Who did what

Member	Code part	Lead of
Rudransh	Part 1: imports, loading CSV, target graph, X/y split; final probability heatmap	Documentation
Purvi	Part 2: feature groups, numeric summary, histograms	PPT
Jathin	Part 3: Seaborn comparisons by Target, binary and categorical analysis	Execution
Divyansh	Part 4: preprocessing, Logistic Regression pipeline, evaluation	Video

AI-use disclosure

- Claude helped us understand basic concepts, including pipelines and Seaborn graphs.
- The slides were enhanced after the first draft.
- AI was not used to generate the dataset or results.
- We ran the code ourselves; all numbers came from our notebook.